

MOOMBA, Australia

WMO: 954810

Lat: 28.0997S

Lon: 140.1956E

Elev: 38

StdP: 100.86

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.1	5.3	-12.1	1.3	24.4	-10.0	1.6	23.6	11.0	19.9	9.7	18.4	2.7	190	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.1	43.0	20.5	41.5	20.1	40.1	19.7	24.8	31.7	23.7	32.0	22.7	32.8	5.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.1	27.6	21.4	16.2	27.3	19.8	14.5	27.2	75.8	31.9	71.1	31.5	67.2	32.5	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.8	1.8	44.9	1.1	1.4	1.0	45.9	0.3	46.7	-0.3	47.5	-1.1	48.5		
(5)			-0.4	28.5	1.2	3.7	-1.3	31.2	-2.0	33.3	-2.7	35.4	-3.6	38.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	31.7	30.4	27.2	22.4	17.3	13.7	13.1	14.9	19.3	23.3	26.6	29.7
	DBStd	7.36	3.52	3.72	3.62	3.49	3.17	2.91	2.66	3.12	3.93	4.11	4.08	3.75
	HDD10.0	4	0	0	0	0	0	2	2	1	0	0	0	0
	HDD18.3	532	0	0	0	7	60	143	164	118	35	6	1	0
	CDD10.0	4539	674	571	532	372	226	113	99	151	278	413	499	612
	CDD18.3	2024	416	337	273	129	27	4	2	10	62	161	249	353
	CDH23.3	29164	6617	5105	3606	1401	259	39	41	157	851	2156	3502	5431
	CDH26.7	17497	4438	3268	2038	596	55	4	5	39	372	1141	2054	3487
Wind	WSAvg	4.7	5.3	5.3	5.1	4.6	4.1	4.0	3.9	4.3	4.6	5.0	5.1	5.2
Precipitation	PrecAvg	175	27	26	4	13	16	11	17	9	11	19	13	10
	PrecMax	345	180	162	34	110	87	43	89	42	113	79	39	100
	PrecMin	45	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	89	41	39	7	23	22	12	23	11	21	20	11	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	43.7	41.1	35.8	30.1	26.9	27.0	30.6	36.4	39.4	43.1	43.9
		MCWB	20.9	21.4	20.2	18.9	16.2	15.6	14.5	15.2	16.4	18.3	19.4	19.8
	2%	DB	43.5	41.8	38.6	34.1	28.2	24.2	24.1	27.2	33.6	37.3	40.4	41.9
		MCWB	20.9	20.7	19.5	17.8	15.5	14.4	13.0	13.6	15.5	17.3	18.5	19.3
	5%	DB	41.6	40.3	37.0	32.2	26.3	21.8	21.8	24.7	31.0	35.3	38.1	40.3
		MCWB	20.5	20.3	19.3	17.0	15.0	13.2	11.9	12.4	14.4	16.8	18.0	19.2
	10%	DB	40.0	38.5	35.3	30.3	24.2	19.8	19.6	22.4	28.3	32.9	35.8	38.4
		MCWB	20.2	20.0	18.5	16.5	14.1	12.1	11.2	11.5	13.8	16.0	17.6	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.8	26.0	24.8	22.1	19.8	17.7	17.6	17.1	19.4	21.6	23.2	25.9
		MCDB	30.9	32.5	28.1	27.4	23.1	21.9	21.7	23.8	24.1	30.8	32.0	31.9
	2%	WB	24.7	24.7	23.2	20.4	17.5	16.2	14.6	15.0	17.7	19.6	21.7	23.1
		MCDB	33.2	31.7	30.4	28.0	23.4	21.1	20.3	24.7	26.3	29.7	30.8	32.7
	5%	WB	23.5	23.6	22.0	19.0	16.2	14.6	13.1	13.5	16.3	18.4	20.7	22.0
		MCDB	33.0	32.3	30.9	28.4	23.4	19.5	19.7	22.6	27.8	30.1	30.3	32.5
	10%	WB	22.4	22.5	20.7	17.6	15.0	13.3	11.9	12.2	14.9	17.2	19.5	20.9
		MCDB	34.2	33.3	31.4	28.3	23.1	18.9	18.5	21.1	26.3	30.0	30.6	33.1
Mean Daily Temperature Range	5% DB	MDBR	13.1	12.6	12.7	12.9	12.3	11.7	12.7	13.8	14.4	14.4	13.8	13.6
		MCDBR	15.1	14.5	14.8	14.9	14.5	13.8	15.7	16.8	18.1	17.6	17.0	15.8
		MCWBR	5.4	5.3	5.9	6.5	6.8	7.0	8.0	8.1	7.9	7.6	6.7	6.2
	5% WB	MCDBR	11.9	10.8	11.4	12.1	11.8	10.8	13.0	14.1	13.9	13.9	13.0	12.1
		MCWBR	5.2	4.8	5.4	5.4	6.3	6.2	7.1	7.3	6.7	7.1	5.9	5.9
Clear-Sky Solar Irradiance	taub	0.372	0.366	0.344	0.327	0.308	0.297	0.300	0.316	0.365	0.362	0.373	0.370	
	taud	2.359	2.398	2.433	2.450	2.465	2.486	2.461	2.396	2.264	2.296	2.317	2.344	
	Ebn at Noon	969	958	949	917	889	877	889	914	911	951	962	974	
	Edh at Noon	133	124	113	101	89	83	88	104	132	136	137	135	
All-Sky Solar Radiation	RadAvg	8.08	7.49	6.54	5.28	4.12	3.55	3.86	4.85	6.07	7.20	7.68	8.02	
	RadStd	0.32	0.49	0.39	0.24	0.19	0.20	0.17	0.22	0.34	0.39	0.42	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page